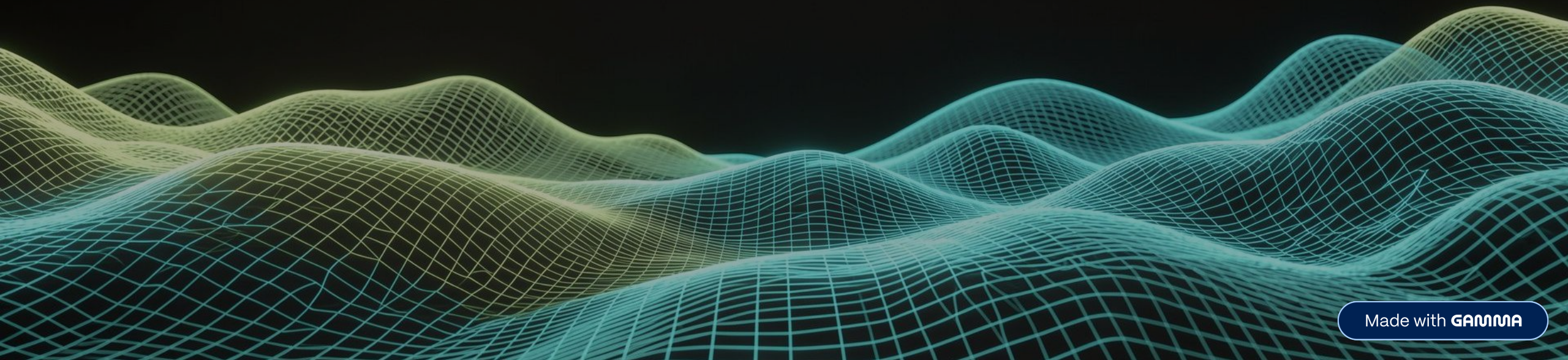
SAFETY INTELLIGENCE PLATFORM

SafeGuard Vision

Industrial safety intelligence from workplace media, PPE detection, incident workflows, and longitudinal analytics.



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01 · VISIBILITY

One operating view for safety leaders

Monitor the safety score, risk score, confirmed violations, worker detections, and surveillance media.

Investigate trend lines and PPE violation distribution from the main dashboard. Act by moving from a detected issue to a report, acknowledgement, or resolution.



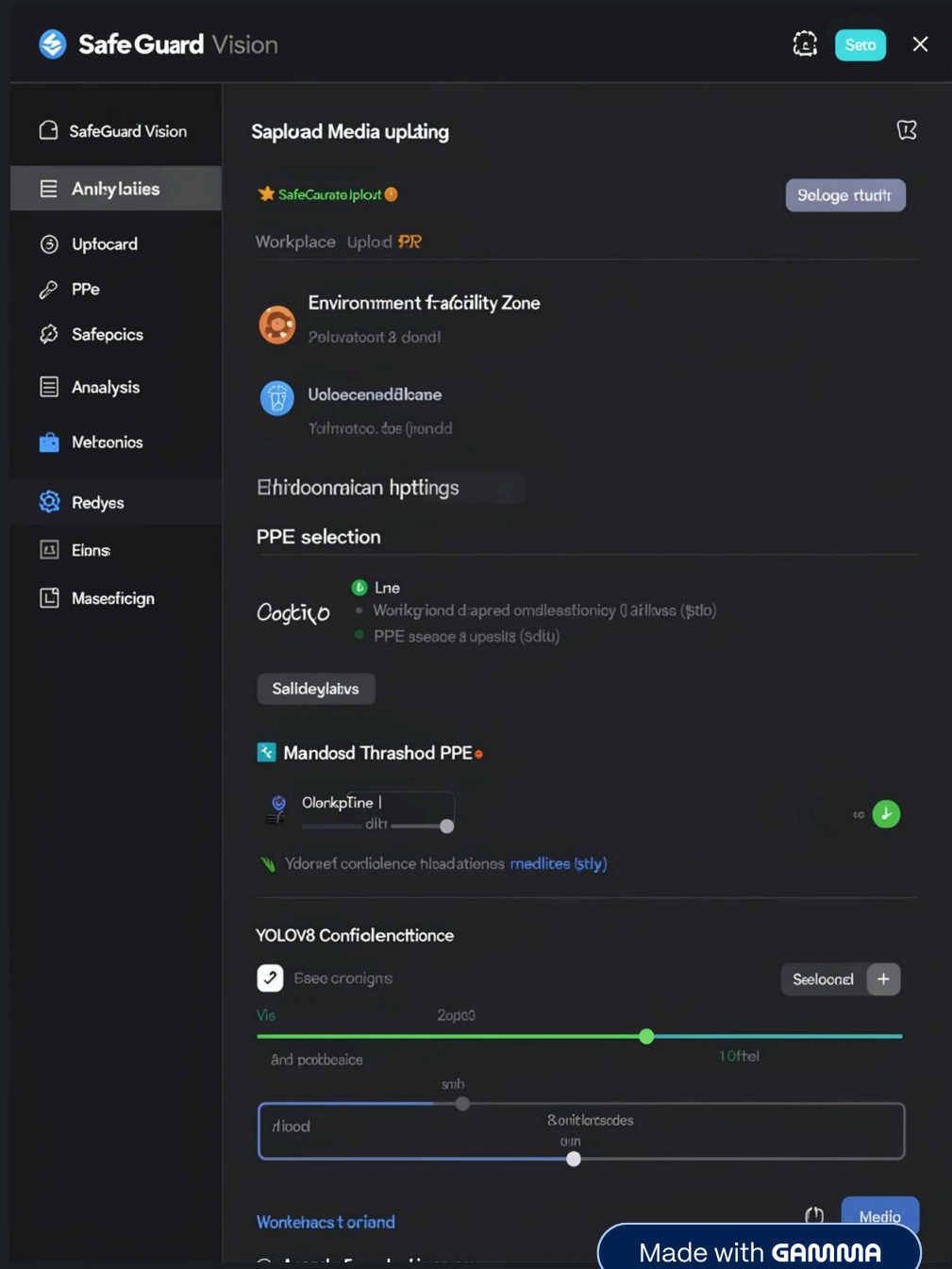
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02 · INGESTION

Upload workplace media and set the rules before analysis

Analysis starts with the evidence and the operating standard that governs it.

1. Bring in factory-floor footage, drone sweeps, handheld photos, or CCTV clips.
2. Choose the workplace environment, facility, zone, and mandatory PPE.
3. Set the YOLOv8 confidence threshold before processing.



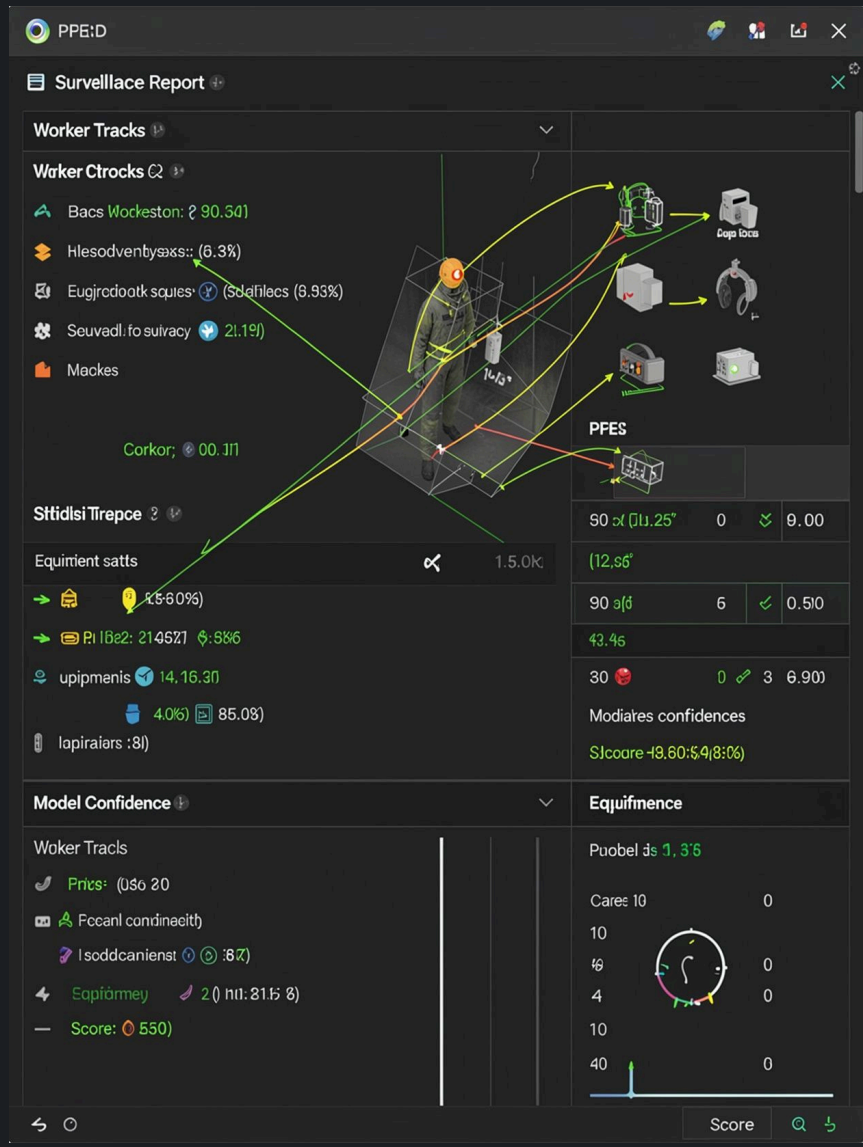
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Every detection becomes an auditable worker-level report

- Show the annotated surveillance feed with worker tracks and PPE overlays.
- Inspect each worker's equipment status and model confidence.
- Explain the score with visible math.

$$2 \text{ workers} \times 5 \text{ PPE items} = 10 \text{ checks}$$

$$100 \times (1 - 20 \div 200) = 90\%$$

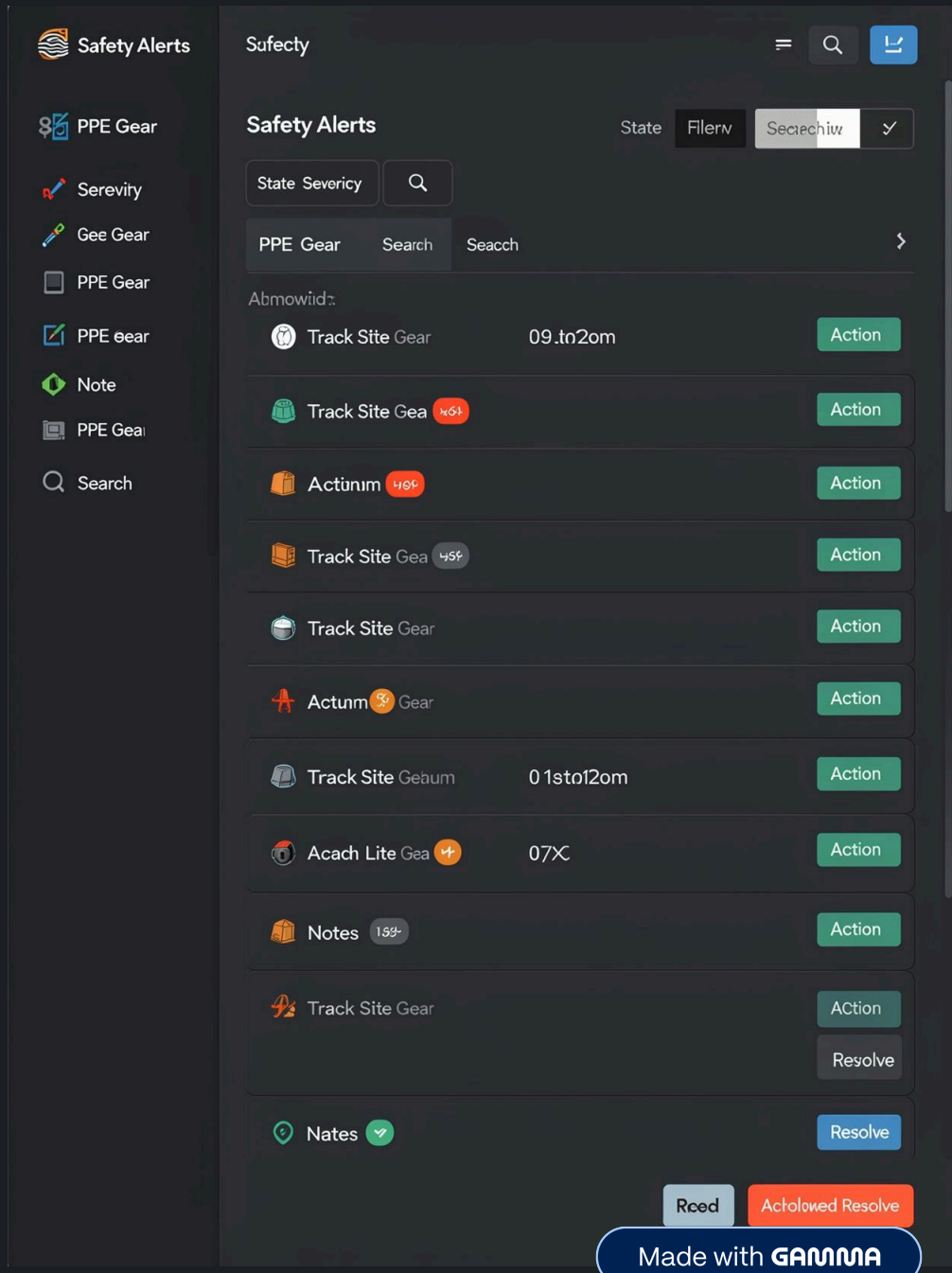


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04 · RESPONSE

Alerts turn non-compliance into an action queue

- Filter by state, severity, PPE gear, or search.
- Review the track, site, gear, timestamp, and action notes.
- Acknowledge or resolve issues directly from the queue.



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05 · ANALYTICS

Longitudinal analytics show where risk is moving

Track daily safety average, risk index, personnel, confirmed violations, and surveillance files.

Read non-compliance density by time of day and historical safety rating across facilities. Use the PPE audit table to see compliance rates and primary failure reasons.

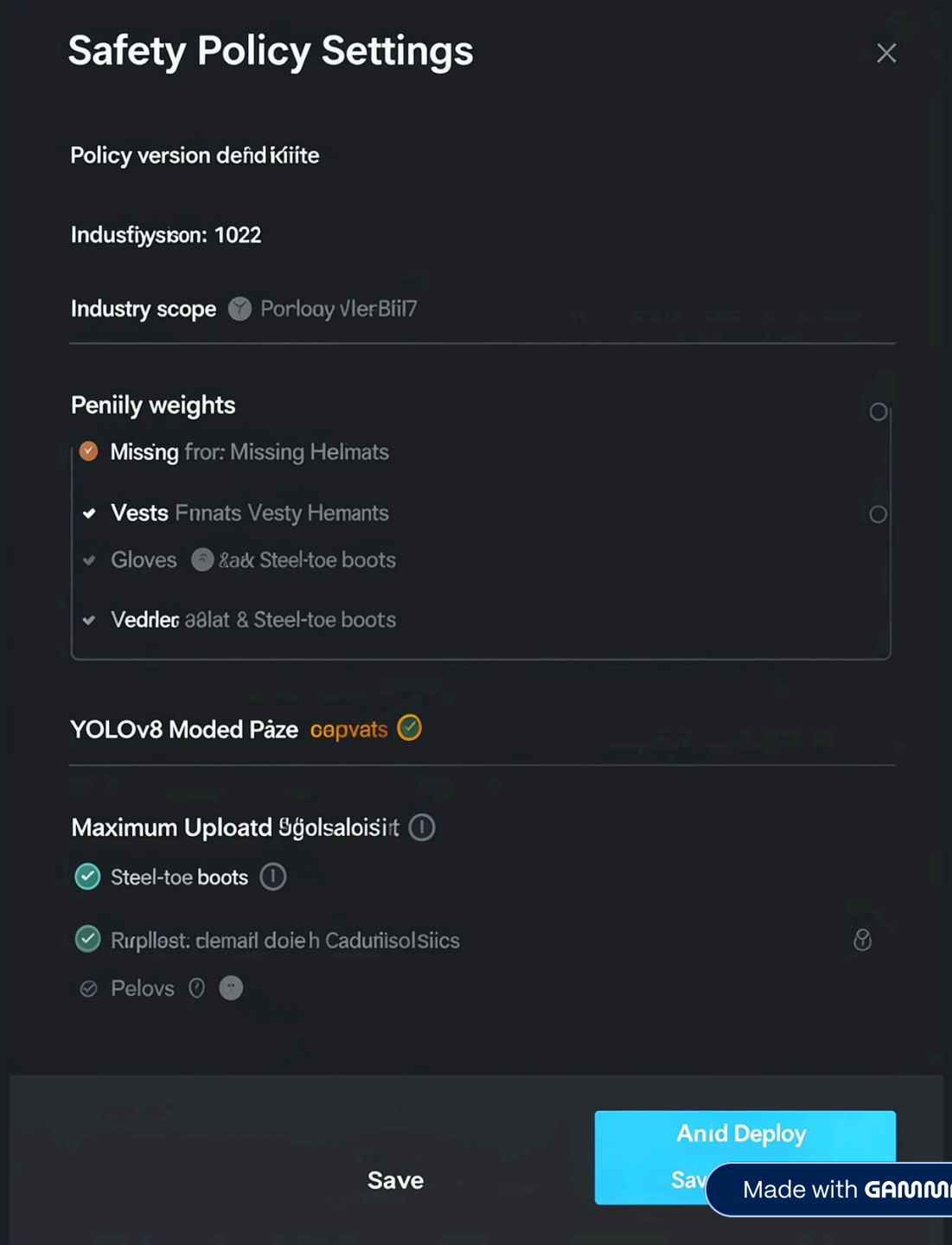


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06 · POLICY

Policy settings make the scoring model transparent

- Identify the policy version and industry scope.
- Assign penalty weights for missing helmets, vests, gloves, and steel-toe boots.
- Configure the YOLOv8 model path and maximum upload size, then save and deploy the safety policy.



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07 · GOVERNANCE

Role-based access keeps the workflow accountable

Chief Safety Officer

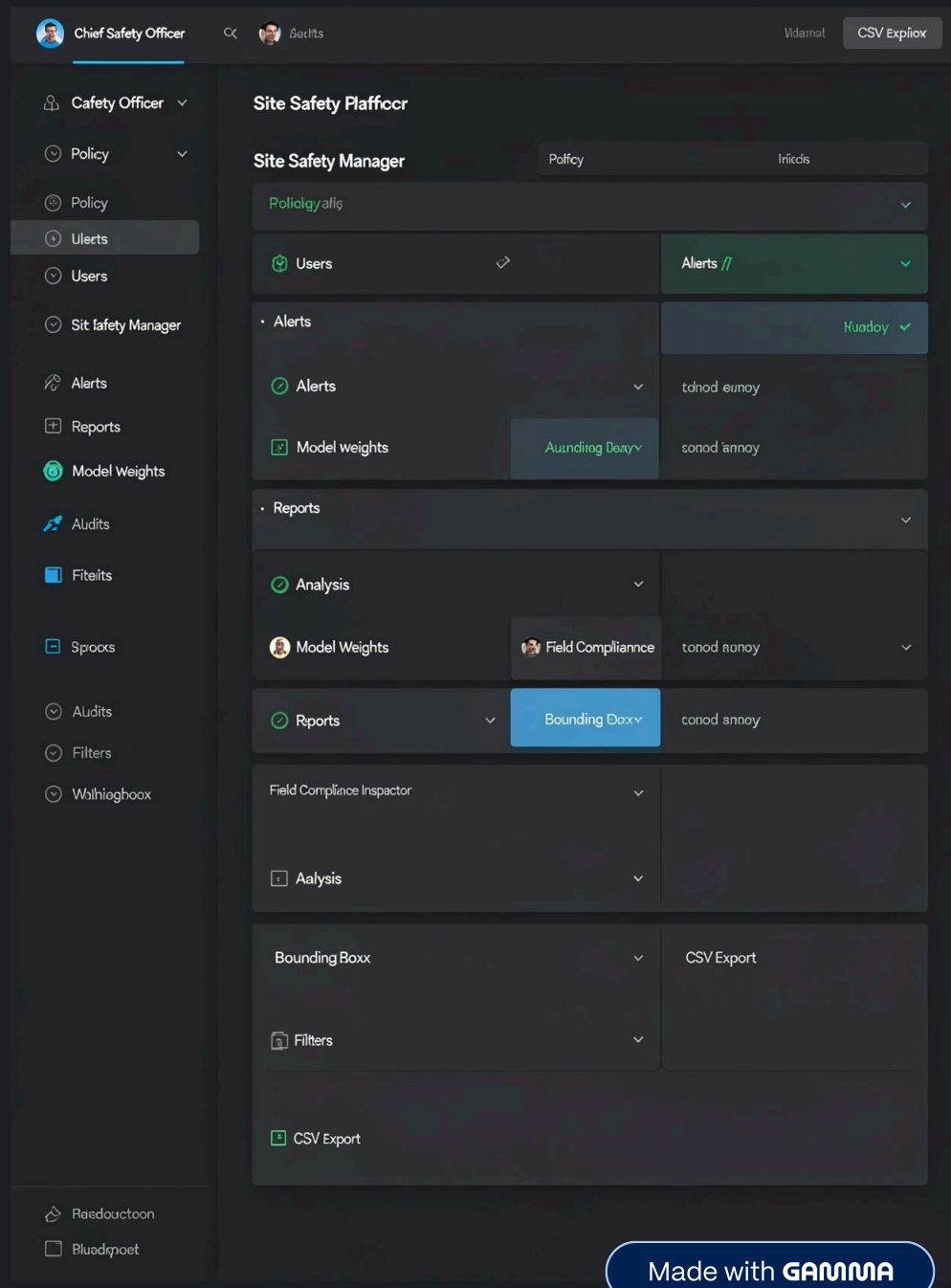
Administrative control over policy, users, alerts, reports, and model weights.

Site Safety Manager

Ingest media, run analysis, acknowledge violations, and export audits.

Field Compliance Inspector

Read-only access to reports, bounding boxes, filters, and CSV export.



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08 · ARCHITECTURE

The platform connects computer vision to operational reporting

- **Computer vision:** Ultralytics YOLOv8 PPE Engine
- **Database:** SQLite with migration support
- **API layer:** Full-Stack Express + Flask YOLOv8 bridge
- **Business intelligence:** Power BI-style analytics and PDF/CSV audit export



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THE OPERATING LOOP

SafeGuard Vision closes the loop

Capture media, apply policy, detect PPE gaps, quantify risk, resolve incidents, and learn from the trend line.

For leaders

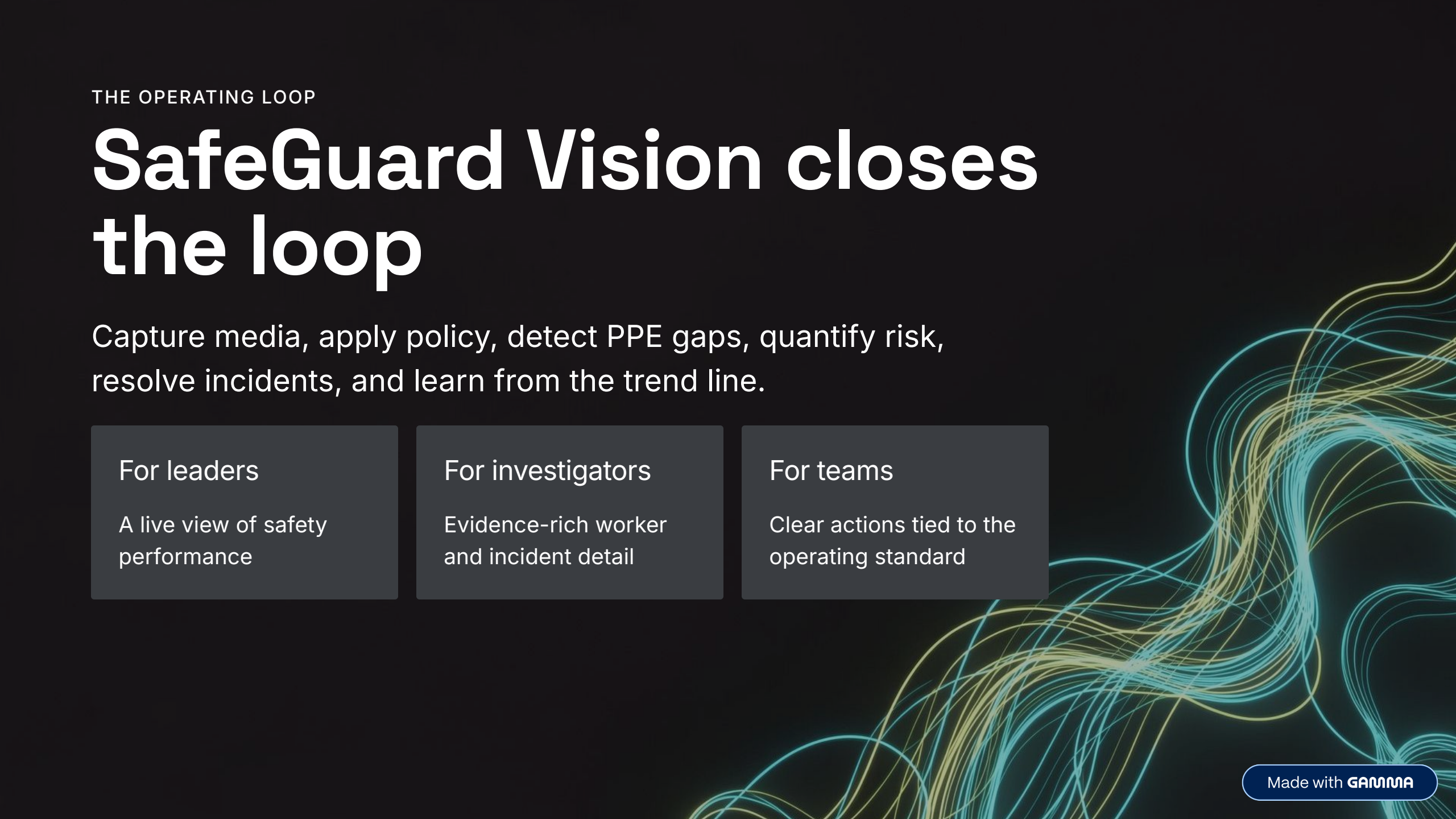
A live view of safety performance

For investigators

Evidence-rich worker and incident detail

For teams

Clear actions tied to the operating standard



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